

Dual pressure surgical air systems

- 4.51 There are cases where, because of system size, a simple single regulation system (that is, directly from a receiver pressure of say, 10 bar, to a line pressure of approximately 8 bar) will not ensure correct flow conditions at the surgical air terminal units. To overcome possible flow problems, a double pressure regulating system can be used.
- 4.52 Such a system will involve a compressed air plant receiver operating at a typical pressure of 13 bar, followed by first-stage pressure regulation to a line pressure of 11 bar. Locally-sited pressure regulators (for example for each operating room) are provided to give the recommended flow and pressure at the terminal unit outlet(s).
- 4.53 If this type of system is installed, for design purposes the maximum allowable pressure drop of 5% should be taken from the plantroom wall to the upstream side of the secondary regulator. The secondary regulator should be adjusted to give 700 kPa at a flow of 350 L/min at the terminal unit outlets(s) and should not allow the static pressure on the upstream side of the terminal unit to rise above 9 bar.

Diversity

- 4.54 Surgical air 700 kPa is only required where surgical tools are to be used. This would typically be orthopaedic and neurosurgery operating rooms, and possibly plaster rooms. For flexibility, and to allow for possible overspill, surgical air should be extended to two to four adjacent operating rooms. It is not required in maternity or ophthalmology operating rooms.

Note

Health Technical Memorandum 2022, Supplement 1 – ‘Dental compressed air and vacuum systems’ allows for the extension of surgical air into dental departments for tool use only. No diversity factor should be applied to the dental service as all dental clinics can be in use simultaneously: the total design flow of the dental department should be added.

- 4.55 Unlike dental departments, the use of surgical tools in an operating procedure takes place for a limited period of time.

Table 19 Typical pressure and flow requirements for surgical tools

Type of tool	Pressure (kPa)	Flow (L/min)
Small air drill	600–700	200
Medullary reaming machine	600–700	350
Oscillating bone saw	600–700	300
Universal drill	600–700	300
Craniotome	620–750	300

System capacity

- 4.56 Unlike respirable equipment, surgical tools are used intermittently, typically for a few seconds, up to a maximum of three minutes. The plant, therefore, should have the capacity to provide the design flow of the pipeline for a maximum period of five minutes in any 15-minute period. The diversified flow is based on the assumption of 350 L/min for the first theatre and a quarter of the remainder – see Table 20.

Terminal units intended for equipment testing

- 4.57 It may be necessary to provide surgical air at 700 kPa in the equipment service workshop for testing purposes. Unless a surgical air 700 kPa pipeline is available nearby, it may be cost-effective

Table 20 Surgical air 700 kPa – design and diversified flows

Department	Design flow for each terminal unit (L/min)	Diversified flow Q (L/min)
Operating room (orthopaedic and neurosurgical operating rooms only): <4 operating rooms >4 operating rooms	350 350	$Q = 350 + [(n - 1)350/2]$ $Q = 350 + [(n - 1)350/4]$
Other departments, eg equipment workshops, fracture clinic	350	$Q = 350$
Equipment service rooms	350	No additional flow required